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Team name on Kaggle leaderboard: Blood Wolf the Pojun

For each of the sections below, your reported test accuracy should approximately match the accuracy reported on Kaggle.

Perceptron

Regarding Perceptron, in the Fashion-MNIST DATASET, first, we kept learning_rate unchanged (learning_rate=0.005), and then adjusted the parameters of n_epochs (10,20,50,75). Then we find that the accuracy is the highest when n_epochs=20. The result is shown in the figure.

learning_rate=0.005			
n_epochs	Training accuracy	Validation accuracy	Test accuracy
10	83.130000	79.730000	79.400000
20	85.842000	81.090000	80.760000
50	85.850000	80.750000	80.620000
75	85.854000	80.750000	80.600000

Then, while keeping n_epochs unchanged (n_epochs=50), we adjusted the value of learning_rate (5, 0.5, 0.05, 0.005, 0.0005). Finally, we found that learning_rate=0.0005, the best effect. The result is shown in the figure.

n_epochs=20			
learning_rate	Training accuracy	Validation accuracy	Test accuracy
5	85.532000	80.770000	80.640000
0.5	85.354000	80.550000	80.620000
0.05	85.424000	80.680000	80.700000
0.005	85.842000	81.090000	80.760000
0.0005	87.454000	82.470000	81.900000

We also adjusted decay_rate. After repeated attempts, we found that for 2-class, when decay_rate is less than 0.95, accuracy gradually increases with the increase of decay_rate. However, when decay_rate is greater than 0.95, accuracy decreases gradually with the increase of decay_rate. For multi-class, we first try the decay_rate of 0.95, and found the validation accuracy is low. So we decreased the decay by 0.05 and got the highest accuracy with 0.80. With this value, the learning rate is approximately 10^(-6) at the end epoch. When decay_rate is lower than 0.80, accuracy begin to decrease gradually.

Similarly, we adjusted the parameters in the same way for the RICE DATASET. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.05 n_epochs = 50 decay_rate = 0.95
Training accuracy:	99.871689
Validation accuracy:	99.807534
Test accuracy:	99.780038

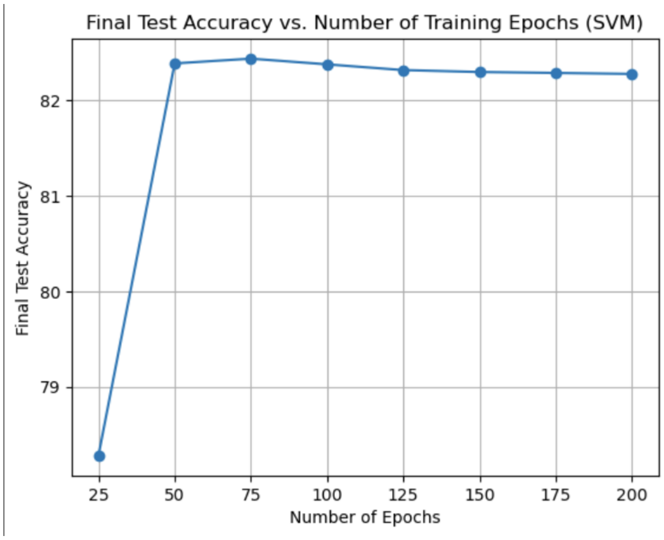
Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.0005 n_epochs = 20 decay_rate = 0.80
Training accuracy:	87.454000
Validation accuracy:	82.470000
Test accuracy:	81.900000

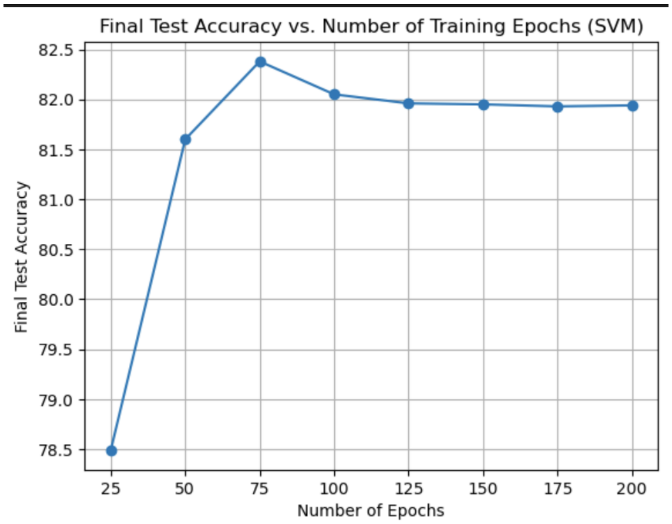
SVM

As for SVM, We first adjusted reg_const from 0.005 to 5.0, with 10 times mulitplication for each step, and found that it worked best when reg_const = 0.5. Then, we use python to draw graphs to predict accuracy. Four groups of pictures were drawn at learning_rate= 0.01, 0.05, 0.5 and 5. Where, the horizontal coordinate of the picture is the number of training rounds, and the ordinate is the corresponding accurancy. As shown in the four graphs below.

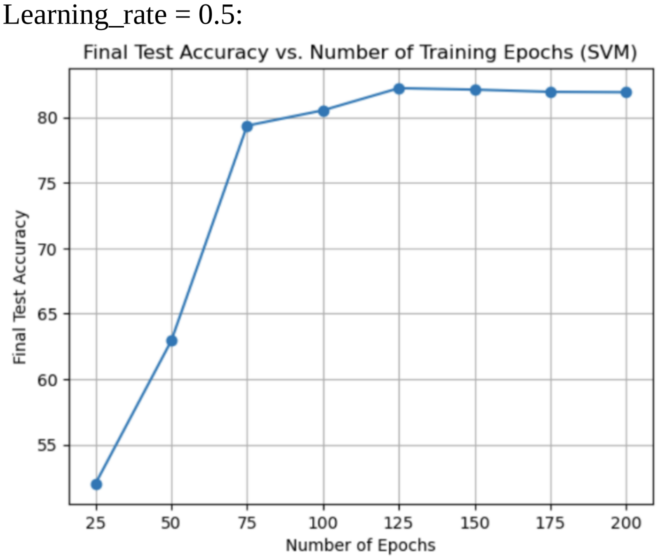
Learning_rate = 0.01:



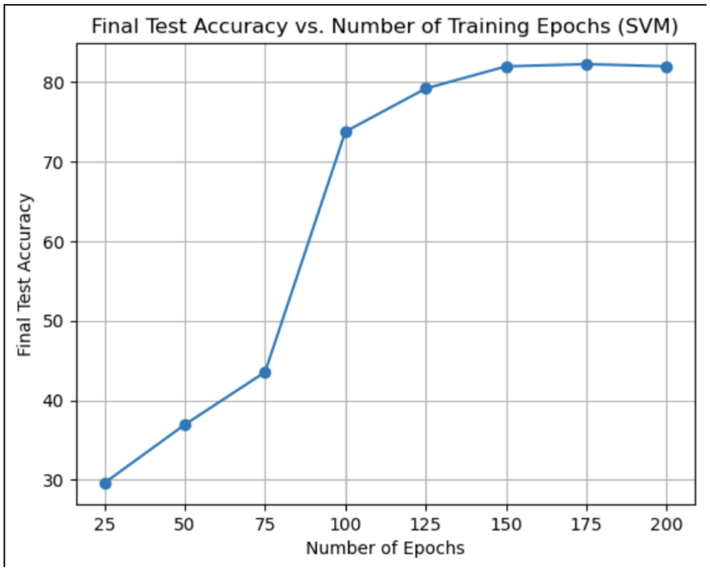
Learning_rate = 0.05



Learning_rate = 0.5:



Learning_rate = 5:



After comparing several groups of pictures, we find that when learning_rate = 0.01, the final effect is best around the 75th round. In addition, we find that the final effect is best when decay_rate = 0.95.

The adjustment for hyperparameters in RICE DATASET were similar, and we found that the regularization constant for this dataset is higher than previous case. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 50 re_const = 1 decay_rate = 0.95
Training accuracy:	97.103840
Validation accuracy:	97.195491
Test accuracy:	97.167996

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.01 n_epochs = 75 re_const = 0.5 decay_rate = 0.95
Training accuracy:	84.590000
Validation accuracy:	82.980000

Test accuracy:	82.440000

Softmax

Once again, describe the hyperparameter tuning you tried for learning rate, number of epochs, and regularization constant. Report the optimal hyperparameter setting you found in the table below. Also report your training, validation, and testing accuracy with your optimal hyperparameter setting.

Regarding the Softmax, in the Fashion-MNIST DATASET, we first kept learning_rate unchanged (learning_rate=0.5) and re_const unchanged (re_const=0.005). Then the n_epochs parameters were adjusted (25,50,75), and it was found that the accuracy was the highest when n_epochs=50. The result is shown in the figure.

lr=0.5	re_const=0.005		
n_epochs	Training accuracy	Validation accuracy	Test accuracy
25	86.376000	85.190000	84.090000
50	86.668000	85.450000	84.310000
75	86.668000	85.420000	84.370000

Then we adjust the value of learning_rate (5, 0.5, 0.05) while keeping n_epochs unchanged (n_epochs=50) and re_const unchanged (re_const=0.005). Finally, we found that learning_rate=0.5, the best effect. The result is shown in the figure.

n_epochs=50	re_const=0.005		
lr	Training accuracy	Validation accuracy	Test accuracy
5	82.182000	81.200000	80.170000
0.5	86.668000	85.450000	84.310000
0.05	84.636000	83.840000	82.920000

After that, while keeping n_epochs unchanged (n_epochs=50) and learning_rate unchanged (learning_rate=0.5), we adjusted re_const numerically (0.5, 0.05, 0.005) and finally found that, This works best when re_const=0.0005. The result is shown in the figure.

n_epochs=50	lr=0.5		
re_const	Training accuracy	Validation accuracy	Test accuracy
0.05	84.924000	84.010000	83.150000

0.005	86.668000	85.450000	84.310000
0.0005	86.894000	85.590000	84.380000

We also adjusted decay_rate. After repeated attempts, we found that when decay_rate is less than 0.95, the accuracy gradually increases with the increase of decay_rate. However, when decay_rate is greater than 0.95, accuracy decreases gradually with the increase of decay_rate.

Similarly, we adjusted the parameters in the same way for the RICE DATASET. Parameters of the RICE DATASET and Fashion-MNIST DATASET were finally adjusted, as shown in the figure below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 100 re_const = 0.05 decay_rate = 0.95
Training accuracy:	99.880854
Validation accuracy:	99.752543
Test accuracy:	99.835029

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate = 0.5 n_epochs = 50 re_const = 0.0005 decay_rate = 0.95
Training accuracy:	86.894000
Validation accuracy:	85.590000
Test accuracy:	84.380000

Logistic

Once again, describe the hyperparameter tuning you tried for learning rate, number of epochs, and threshold. Report the optimal hyperparameter setting you found in the table below. Also report your training, validation, and testing accuracy with your optimal hyperparameter setting.

Regarding hyperparameter tuning, considering the intrinsic properties of Logistic Regression, the threshold at 0.5 serves as the symmetric center of the function, making it the decision boundary between the two classes. Therefore, this parameter does not need to be adjusted and should remain at threshold = 0.5.

For tuning learning rate and number of epochs, our group initially set n_epochs to an small value (n_epochs = 150) while varying the learning rate by factors of 10^(-1) to determine its optimal range. The specific results are shown in the figure.

n_epochs=150	threshold=0.5		
learning_rate	Training accuracy	Validation accuracy	Test accuracy
5	99.972505	90.266703	89.991751
0.5	99.899184	98.817707	99.010173
0.05	99.798369	93.841078	94.088535
0.005	99.569242	48.144075	49.821281

Subsequently, we found that when learning_rate = 0.5, the performance was relatively good. Keeping learning_rate = 0.5 fixed, we then increased n_epochs and observed that when n_epochs was below 150, increasing n_epochs led to improved performance. However, beyond n_epochs = 150, the accuracy hardly improved any further, as illustrated in the figure.

learning_rate=0.5	threshold=0.5		
n_epochs	Training accuracy	Validation accuracy	Test accuracy
50	99.880854	98.790212	98.955183
100	99.880854	98.872697	99.037668
150	99.899184	99.037668	99.120154
200	99.917514	98.597745	98.900192

Finally, we explored the effect of decay_rate. We found that as decay_rate decreased, the accuracy also dropped significantly, as shown in the figure.

learning_rate=0.5	threshold=0.5	n_epochs=150	
decay_rate	Training accuracy	Validation accuracy	Test accuracy
1	99.899184	99.037668	99.120154
0.99	99.880854	98.817707	99.010173
0.95	99.798369	92.768765	92.466318
0.9	99.761708	82.430575	82.815507

In conclusion, the optimal hyperparameters were determined as learning_rate = 0.5, threshold = 0.5, n_epochs = 150, and no decay_rate. The final results are summarized in the table below.

RICE DATASET

Optimal hyperparameters:	learning_rate = 5 n_epochs = 150 threshold = 0.5
Training accuracy:	99.972505
Validation accuracy:	99.917514
Test accuracy:	99.972505